

Probabilistic Graphical Models

CSE 5830

Foundation

Probability and Graphs

Lecture 2

Sample space and Events

- Probability: a degree of confidence
- **Space** of possible **outcomes** $\rightarrow \Omega$
- There is a set of measurable **events** $\mathcal{S}$ to which we want to assign probabilities

Probability theory

Event space satisfy three basic properties

- Contains empty event and trivial event
- Closed under union
- Closed under complementation

Probability distribution

A probability distribution P over (S, Ω) is a mapping from events in S to real value that satisfy:

- $P(\alpha) \geq 0$ for all $\alpha \in S$
- $P(\Omega) = 1$
- if $\alpha, \beta \in S$ and $\alpha \cap \beta = \emptyset \Rightarrow P(\alpha \cup \beta) = P(\alpha) + P(\beta)$

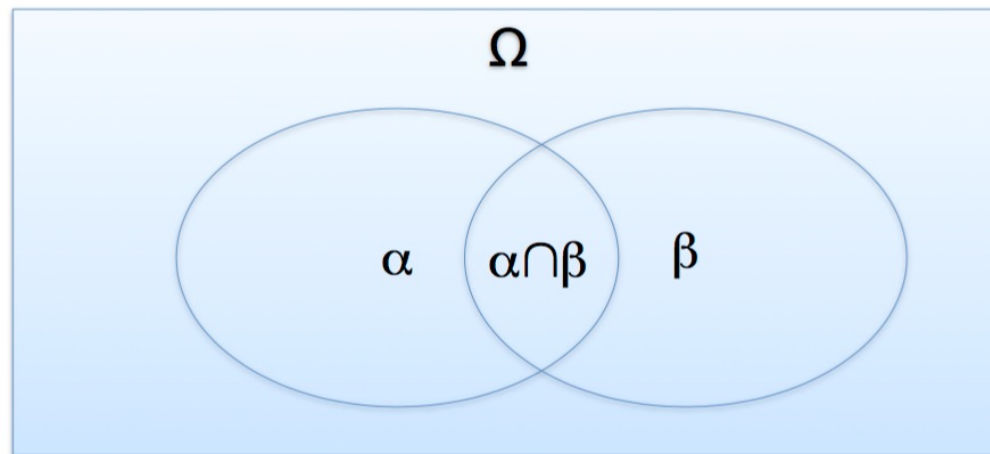
Two Interpretations of Probability

- Frequentists
 - $P(\alpha)$ is the frequency of α in the limit
- Subjective
 - $P(\alpha)$ is the degree of belief that α will happen

Conditional Probabilities

- After learning that α is true, how do we feel about β ?

$$P(\beta \mid \alpha) = \frac{P(\alpha \cap \beta)}{P(\alpha)}$$



Chain Rule

- Chain rule

$$P(\beta \mid \alpha) = \frac{P(\alpha \cap \beta)}{P(\alpha)} \quad \longrightarrow \quad P(\alpha \cap \beta) = P(\alpha)P(\beta \mid \alpha)$$

In general for events $\alpha_1 \dots \alpha_k$

$$P(\alpha_1 \cap \dots \cap \alpha_k) = P(\alpha_1)P(\alpha_2 \mid \alpha_1) \cdots P(\alpha_k \mid \alpha_1 \cap \dots \cap \alpha_{k-1})$$

Factorization

Bayes Rule

$$P(\beta \mid \alpha) = \frac{P(\alpha \cap \beta)}{P(\alpha)}$$

$$P(\alpha \mid \beta)P(\beta) = P(\alpha \cap \beta) = P(\beta \mid \alpha)P(\alpha)$$

$$P(\alpha \mid \beta) = \frac{P(\beta \mid \alpha)P(\alpha)}{P(\beta)}$$

Bayes Rule

The diagram illustrates the components of Bayes' Rule. The equation is
$$P(\alpha | \beta) = \frac{P(\beta | \alpha)P(\alpha)}{P(\beta)}$$
. Four blue arrows point from labels to parts of the equation: 'likelihood' points to $P(\beta | \alpha)$, 'prior' points to $P(\alpha)$, 'posterior' points to $P(\alpha | \beta)$, and 'normalization constant' points to $P(\beta)$.

likelihood

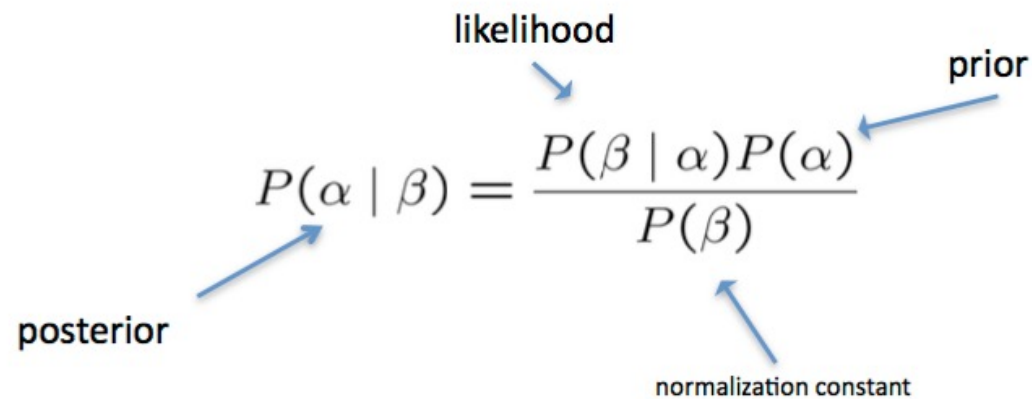
prior

posterior

normalization constant

$$P(\alpha | \beta) = \frac{P(\beta | \alpha)P(\alpha)}{P(\beta)}$$

More general form of Bayes Rule



A diagram showing the Bayes' Rule equation $P(\alpha | \beta) = \frac{P(\beta | \alpha)P(\alpha)}{P(\beta)}$. Four blue arrows point from text labels to parts of the equation: 'likelihood' points to $P(\beta | \alpha)$, 'prior' points to $P(\alpha)$, 'posterior' points to $P(\alpha | \beta)$, and 'normalization constant' points to $P(\beta)$.

$$P(\alpha | \beta) = \frac{P(\beta | \alpha)P(\alpha)}{P(\beta)}$$

likelihood

prior

posterior

normalization constant

$$P(\alpha | \beta \cap \gamma) = \frac{P(\beta | \alpha \cap \gamma)P(\alpha | \gamma)}{P(\beta | \gamma)} \quad \text{where } \gamma \text{ is an "external event"}$$

Random variables

- We tend to group events by **attributes** of outcomes
 - Event: a set of outcomes
- **Random variables** formalize attributes
 - A random variable is a function that associates each outcome with a value
e.g. :“Grade = A” is shorthand for the set of events:
$$\{\omega \in \Omega : f_{\text{Grade}}(\omega) = A$$

A way of reporting an attribute of the outcome
 - A random variable can be
 - Categorical (discrete)
 - Continues (real value)

Joint distribution of random variables

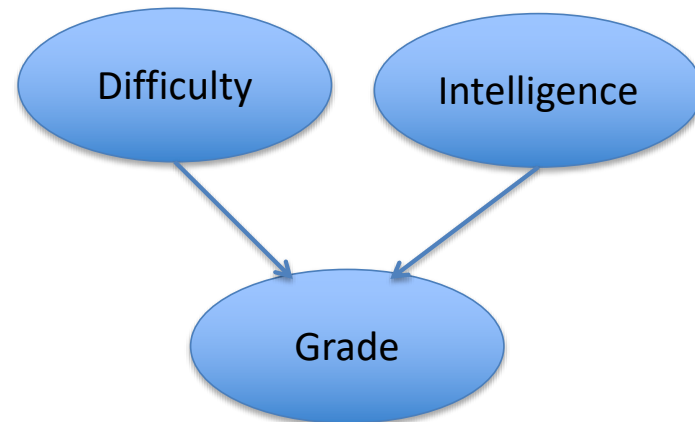
- Joint distribution: $P(X_1, \dots, X_n)$
- For categorical random variable the most practical way to show probabilities is Tabular Representation (Probability Tables)

Example

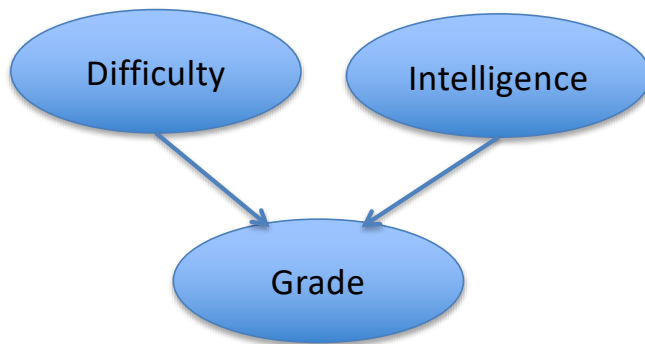
Joint distribution for categorical data

Probability queries

- Intelligence (I)
 i^0 (low), i^1 (high),
- Difficulty (D)
 d^0 (easy), d^1 (hard)
- Grade (G)
 g^1 (A), g^2 (B), g^3 (C)



Joint distribution



- Intelligence (I)
 i^0 (low), i^1 (high),
- Difficulty (D)
 d^0 (easy), d^1 (hard)
- Grade (G)
 g^1 (A), g^2 (B), g^3 (C)

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^0	g^2	0.168
i^0	d^0	g^3	0.126
i^0	d^1	g^1	0.009
i^0	d^1	g^2	0.045
i^0	d^1	g^3	0.126
i^1	d^0	g^1	0.252
i^1	d^0	g^2	0.0224
i^1	d^0	g^3	0.0056
i^1	d^1	g^1	0.06
i^1	d^1	g^2	0.036
i^1	d^1	g^3	0.024

Courtesy to Daphne Koller's slides

Marginal distribution

- For a Joint distribution: $P(X_1, \dots, X_n)$ we can define marginal distribution
- $P(X_1)$: marginal distribution over random variable X_1
 - Subset of S that can be described with the random variable X_1

$$P(X_1) = \sum_{X_2, \dots, X_n} P(X_1, \dots, X_n)$$

Marginal distribution

$P(I,D,G)$

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^0	g^2	0.168
i^0	d^0	g^3	0.126
i^0	d^1	g^1	0.009
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i^1	d^0	g^3	0.0056
i^1	d^1	g^1	0.06
i^1	d^1	g^2	0.036
i^1	d^1	g^3	0.024

$P(I,D)$

I	D	Prob.
i^0	d^0	0.420
i^0	d^1	0.180
i^1	d^0	0.280
i^1	d^1	0.120

Conditional Probability for Random Variables

$$P(X|Y) = \frac{P(X,Y)}{P(Y)}$$

Bayes rule

Chain rule

Example Conditioning

Conditioning on g^1

$$P(D, I \mid G = g^1)$$

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^0	g^2	0.168
i^0	d^0	g^3	0.126
i^0	d^1	g^1	0.009
i^0	d^1	g^2	0.045
i^0	d^1	g^3	0.126
i^1	d^0	g^1	0.252
i^1	d^0	g^2	0.0224
i^1	d^0	g^3	0.0056
i^1	d^1	g^1	0.06
i^1	d^1	g^2	0.036
i^1	d^1	g^3	0.024

Example

Conditioning: reduction

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^1	g^1	0.009
i^1	d^0	g^1	0.252
i^1	d^1	g^1	0.06

Example

Conditioning: renormalization

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^1	g^1	0.009
i^1	d^0	g^1	0.252
i^1	d^1	g^1	0.06

$P(I,D,g^1)$

0.447



I	D	Prob.
i^0	d^0	0.282
i^0	d^1	0.02
i^1	d^0	0.564
i^1	d^1	0.134

$P(I,D|g^1)$

Example

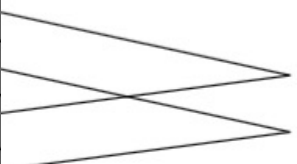
Marginalization

Marginalize I

$$\sum_i P(D, I | G = g^1)$$

$$P(D, I | G = g^1)$$

I	D	Prob.
i^0	d^0	0.282
i^0	d^1	0.02
i^1	d^0	0.564
i^1	d^1	0.134



D	Prob.
d^0	0.846
d^1	0.154

Independence

- α and β are **independent** if

$$P(\beta|\alpha) = P(\beta)$$

- α and β are independent, **written**: $\alpha \perp \beta$

$$P : (\alpha \perp \beta)$$

- **Proposition**: α and β are **independent** if and only if

$$P(\alpha \cap \beta) = P(\alpha) P(\beta)$$

Conditional Independence

- Independence is rarely true.
- α and β are **conditionally independent** given γ if

$$P(\beta \mid \alpha \cap \gamma) = P(\beta \mid \gamma)$$

$$P : (\alpha \perp \beta \mid \gamma)$$

- **Proposition:** $P : (\alpha \perp \beta \mid \gamma)$ if and only if

$$P(\alpha \cap \beta \mid \gamma) = P(\alpha \mid \gamma) P(\beta \mid \gamma)$$

Conditional Independence of Random Variables

- Sets of variables X, Y, Z
- X is independent of Y given Z if

$$P : (X=x \perp Y=y \mid Z=z) = P(X=x \mid Z=z) P(Y=y \mid Z=z) \\ \forall x \in \text{Val}(X), y \in \text{Val}(Y), z \in \text{Val}(Z)$$

- Shorthand:
 - Conditional independence: $P: (X \perp Y \mid Z)$
 - For $P: (X \perp Y \mid \emptyset)$, write $P: (X \perp Y)$
- **Proposition:** P satisfies $(X \perp Y \mid Z)$ if and only if

$$P(X, Y \mid Z) = P(X \mid Z) P(Y \mid Z)$$

Properties of Independence

- Symmetry:

$$(X \perp Y \mid Z) \Rightarrow (Y \perp X \mid Z)$$

- Decomposition:

$$(X \perp Y, W \mid Z) \Rightarrow (X \perp Y \mid Z)$$

- Weak union:

$$(X \perp Y, W \mid Z) \Rightarrow (X \perp Y \mid Z, W)$$

- Contraction:

$$(X \perp W \mid Y, Z) \wedge (X \perp Y \mid Z) \Rightarrow (X \perp Y, W \mid Z)$$

- Intersection:

$$(X \perp W \mid Y, Z) \wedge (X \perp Y \mid W, Z) \Rightarrow (X \perp Y, W \mid Z)$$

– Only for positive distributions: $P(\alpha) > 0, \forall \alpha, \alpha \neq \emptyset$

Continuous Spaces

- Probabilities over continuous random variables with real values.
 - Probability density function (pdf)

$$P : \mathcal{R} \rightarrow \mathcal{R}$$

$$P(X \leq a) = \int_{-\infty}^a P(x) dx$$

- For continuous random variable we mostly use closed form parametric joint probability density function
 - multivariate Gaussian

Expectation and variance

- Expectation

$$E_p[X] = \sum_x x.P(x)$$

$$E_p[X] = \int x.P(x)$$

- Variance

$$Var[X] = E_p[(X - E_p[X])^2]$$

$$Var[X] = E_p[X^2] - E_p[X]^2$$

Expectation and variance

- Expectation

- $E[a.X + b] = aE[X] + b$

- $E[X + Y] = E[X] + E[Y]$

Even when X,Y are not independent

- $E[X.Y] = E[X].E[Y]$

When X,Y are independent

- $E_p[X|y] = \sum_x xP(x|y)$

Conditional expectation

- Variance

- $Var[X + Y] = Var[X] + Var[Y]$

When X,Y are independent

- $Var[aX] = a^2Var[X]$

- Standard deviation : $\sigma_x = \sqrt{Var[X]}$

Graph

- Representation of a probability distribution using a graph as a data structure
- A graph consists of a set of nodes and edges

$$X_i \rightarrow X_{i+1}$$

$$X_i - X_{i+1}$$

$$X_i \Leftrightarrow X_{i+1}$$

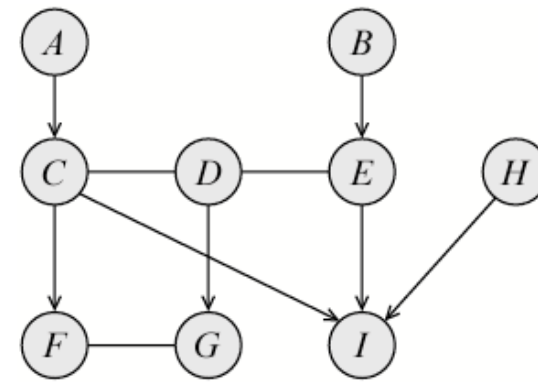
Child, Parent, neighbor nodes

$Pa\ X$

$Ch\ X$

$Nb\ X$

$$Boundary\ X = Pa\ X \cup Nb\ X$$



Partially directed graph

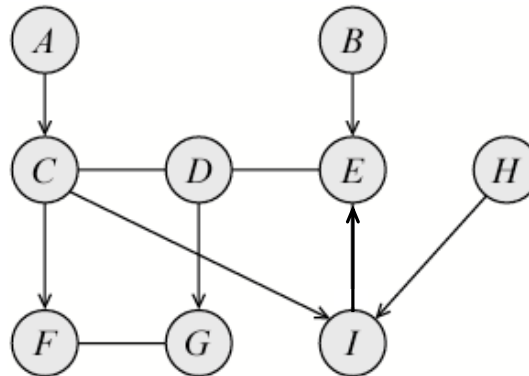
Degree of a graph

- The degree of node X is the number of edges in which it participate
- Degree of a graph is the maximal degree of a node in the graph
- A subgraph is part of a graph that is associated with a particular subset of nodes
- A subgraph is complete if every two nodes are connected by some edge

Paths

Longer range connection

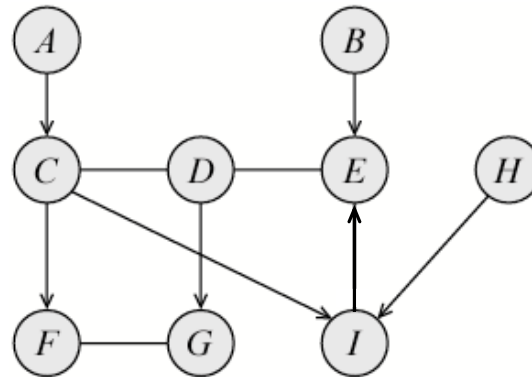
- $X_1, \dots, X_k$ form a path in the graph $G = (X, E)$ if for every $i = 1, \dots, k-1$, we have $X_i \rightarrow X_{i+1}$ or $X_i \dashrightarrow X_{i+1}$
 - A path is directed if at least for one i we have $X_i \rightarrow X_{i+1}$
 - A path is a walk with no repeated node



Trails

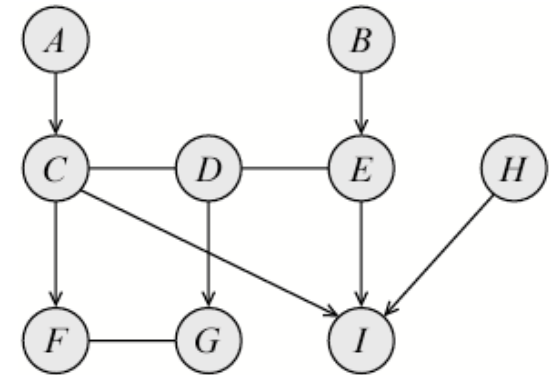
Longer range connection

- $X_1, \dots, X_k$ form a trail in the graph $G = (X, E)$ if for every $i = 1, \dots, k-1$, we have $X_i \rightleftharpoons X_{i+1}$
 - A graph is connected if for every X_i, X_j , there is a trail between X_i, X_j
 - A trail is a walk with no repeated edge, but nodes may repeat



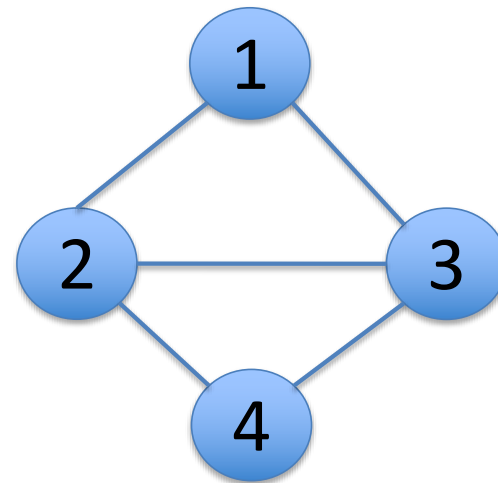
Terminologies

- Ancestor, Descendant
 - X is an ancestor of Y , and Y is a descendant of X if for a directed path:
 $X_1, \dots, X_k \rightarrow X_1 = X, X_k = Y$
- Topological ordering
 - $X_1, \dots, X_k, X_i \rightarrow X_j$ then $i < j$
- Cyclic
 - Path: $X_1, \dots, X_k$, then $X_1 = X_k$
 - No cycle in the graph $\rightarrow$ acyclic
- DAG
 - Directed Acyclic Graph $\rightarrow$ Bayesian network
- PDAG
 - Partially directed acyclic graph $\rightarrow$ undirected edges $\rightarrow$ chain graph



Numerically encoding graphs

- Edge list
- Adjacency matrix



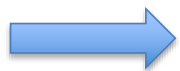
Why Bayesian Networks?

Goal:

Represent a joint distribution P over some set of random variables $\mathcal{X} = \{X_1, \dots, X_n\}$

The explicit representation is not practical

- Computationally, very expensive (memory)
 - E.g. For n binary random variables we need $2^n - 1$ parameter
- Cognitively, can not be handled
- Statistically, not enough samples to estimate parameters



Exploiting independence properties

Exploiting independence properties

- Representation of independence properties of the distribution
- Use of alternative parameterization that allows exploiting of these independencies

Independent random variable

- Without considering independency for n binary random variables we need $2^n - 1$ parameters to represent the distribution
- n independent random variables: $X_1, \dots, X_n$

$$- X_i \perp X_j \Rightarrow P(X_1, \dots, X_n) = P(X_1)P(X_2) \cdots P(X_n)$$

Parameterization

- We need n parameters to represent distribution
- Example tossing coin, θ_i :prob(head) for coin i

Conditional parameterization

- In real word problems, random variables are not independent
- Conditional parameterization provides a basis for compact representation of complex distribution

Example: probability of intelligence and SAT score of a student
Assuming binary random variables I (intelligence) and S (student)
 $P(S,I)$

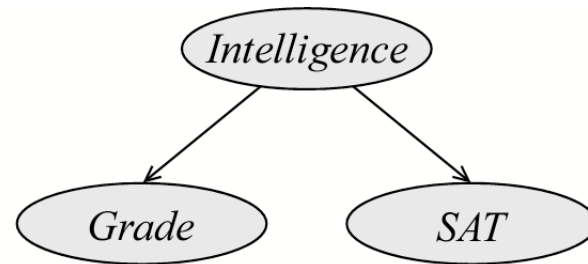


S depends on I or
 I causes S

Conditional independency $\rightarrow$ causality $\rightarrow$ Bayesian Network

Example of simple BN

- Intelligence (I)
 i^0 (low), i^1 (high),
- SAT (S)
 s^0 (low), s^1 (high)
- Grade (G)
 g^1 (A), g^2 (B), g^3 (C)



$$P(I, S, G)$$

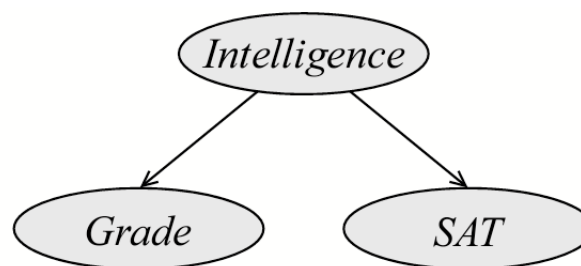
$$P \models (S \perp G \mid I)$$

Compact specification of the joint distribution

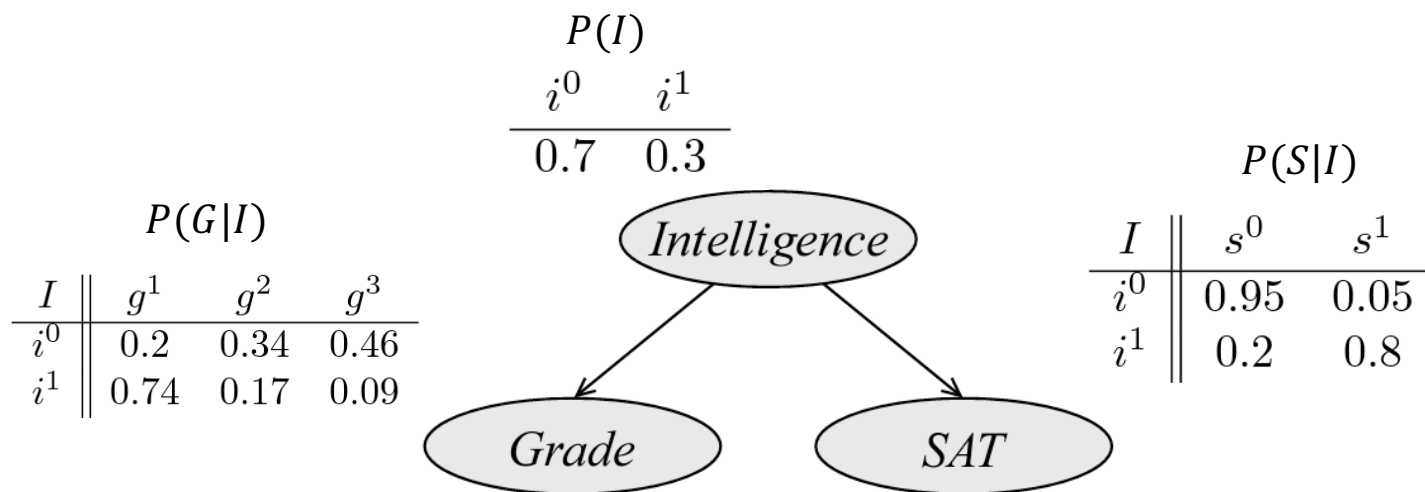
$$P \models (S \perp G \mid I)$$

$$P(I, S, G) = P(S, G \mid I)P(I)$$

$$P(S, G \mid I) = P(S \mid I)P(G \mid I)$$

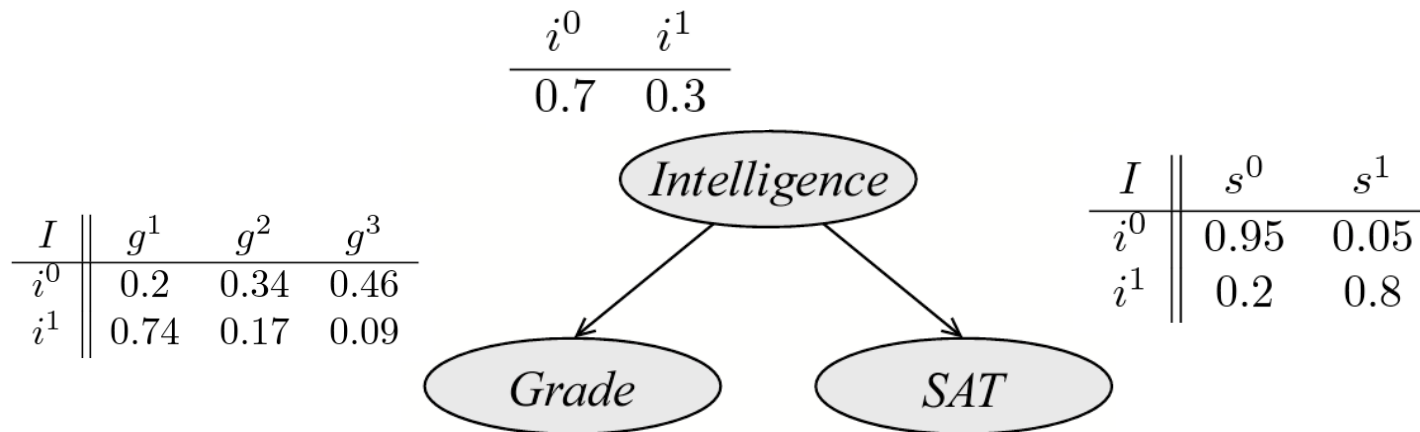


$$P(I, S, G) = P(S \mid I)P(G \mid I)P(I)$$



$$P(i^1, s^1, g^2)$$

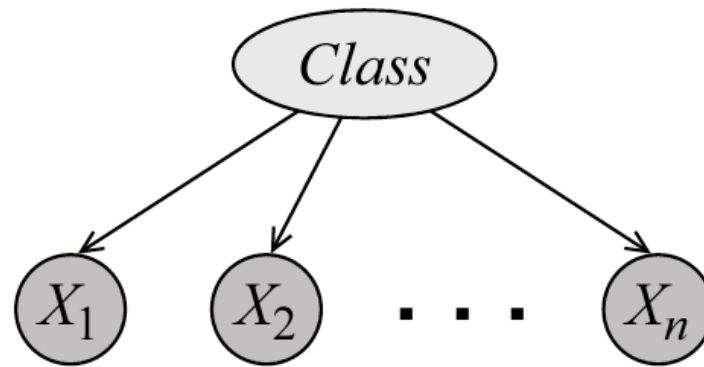
Simple Naïve Bayes model



$$\begin{aligned}
 P(i^1, s^1, g^2) &= P(i^1)P(s^1 | i^1)P(g^2 | i^1) \\
 &= 0.3 \cdot 0.8 \cdot 0.17 = 0.0408.
 \end{aligned}$$

How many parameters we need using the conditional parameterization?
 How many parameters we need using the joint probability?

The Naïve Bayes Model



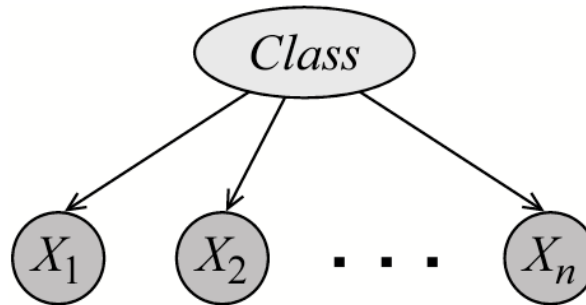
Observed features $X_1, \dots, X_n$ are conditionally independent

$$(X_i \perp \mathbf{X}_{-i} \mid C) \quad \text{for all } i, \quad \text{where } \mathbf{X}_{-i} = \{X_1, \dots, X_n\} - \{X_i\}$$

Model factorization

$$P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i \mid C)$$

The Naïve Bayes Model for Classification



$$\frac{P(C = c^1 \mid x_1, \dots, x_n)}{P(C = c^2 \mid x_1, \dots, x_n)} = \frac{P(C = c^1)}{P(C = c^2)} \prod_{i=1}^n \frac{P(x_i \mid C = c^1)}{P(x_i \mid C = c^2)};$$